

VADUZ, Liechtenstein

WMO: 069900

Lat: 47.1275N

Lon: 9.5175E

Elev: 1519

StdP: 13.91

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.3	19.2	8.2	8.8	22.4	12.0	10.7	23.6	25.2	54.3	20.4	52.9	2.9	330	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	86.4	68.0	83.2	66.8	80.2	65.6	70.0	82.8	68.3	80.3	66.7	77.7	7.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	99.1	76.3	63.8	93.8	74.2	62.5	89.6	72.5	34.8	82.9	33.4	80.5	32.1	78.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	15.8	11.9	DB	11.2	91.4	4.9	3.1	7.7	93.6	4.8	95.4	2.1	97.1	-1.5	99.4
			WB	10.5	73.5	4.9	1.8	7.0	74.8	4.1	75.8	1.4	76.8	-2.2	78.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	34.7	37.5	44.9	51.6	59.1	64.9	67.8	67.2	59.7	52.9	43.0	36.2	(6)
(7)		DBStd	13.96	8.76	8.79	8.02	8.02	7.10	7.41	6.28	6.26	6.37	7.19	8.38	8.18	(7)
(8)		HDD50	1846	475	355	199	77	9	1	0	0	4	52	241	433	(8)
(9)		HDD65	5247	938	769	624	405	210	93	43	49	183	378	662	893	(9)
(10)		CDD50	2466	2	6	40	125	290	448	550	533	294	143	30	5	(10)
(11)		CDD65	394	0	0	0	4	26	91	129	117	22	4	1	0	(11)
(12)		CDH74	2699	0	0	0	20	196	701	934	728	103	17	0	0	(12)
(13)		CDH80	693	0	0	0	1	31	191	269	193	7	1	0	0	(13)
(14)	Wind	WSAvg	4.5	4.8	5.1	5.2	5.6	4.6	4.2	4.0	3.7	3.7	4.1	4.5	4.8	(14)
(15)	Precipitation	PrecAvg	50.0	2.9	3.0	3.3	3.4	4.6	5.6	6.0	6.3	4.4	3.4	3.7	3.4	(15)
(16)		PrecMax	67.7	6.8	9.6	7.6	6.2	10.1	9.3	10.7	11.4	7.0	8.0	9.6	8.7	(16)
(17)		PrecMin	38.3	0.2	0.9	1.1	0.9	0.9	2.4	2.7	1.4	1.8	0.2	0.0	0.3	(17)
(18)		PrecStd	6.1	1.8	2.0	1.7	1.4	2.1	1.9	2.0	2.2	1.5	1.7	2.0	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	62.0	68.9	76.7	84.1	89.7	90.8	89.8	80.9	76.1	69.1	64.1	(19)
(20)		0.4%	MCWB	46.2	46.3	51.2	56.7	62.8	68.5	69.2	69.2	63.7	57.3	52.5	48.8	(20)
(21)		2%	DB	55.3	57.5	64.6	72.8	79.2	85.1	86.4	85.0	77.1	71.5	65.6	57.9	(21)
(22)		2%	MCWB	42.9	44.0	49.6	55.0	60.9	67.0	68.1	68.7	63.4	56.5	50.4	44.8	(22)
(23)		5%	DB	50.8	53.9	60.9	69.1	75.4	81.6	82.7	81.4	73.5	67.6	61.2	51.1	(23)
(24)		5%	MCWB	41.6	42.4	47.9	53.2	59.0	65.9	67.3	67.3	61.5	55.6	49.0	42.7	(24)
(25)		10%	DB	46.6	49.6	57.3	65.3	71.8	78.0	79.3	77.9	70.1	63.9	54.7	47.0	(25)
(26)		10%	MCWB	40.4	41.1	46.3	51.4	57.9	64.2	65.8	65.9	60.5	54.3	47.2	41.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	47.7	53.4	58.8	65.8	71.2	72.2	72.0	67.4	61.1	53.6	49.0	(27)
(28)		0.4%	MCDWB	55.5	57.8	65.9	73.9	79.8	85.8	86.5	85.0	77.1	68.5	65.0	61.9	(28)
(29)		2%	WB	44.6	45.5	50.8	56.3	62.7	68.7	69.7	70.0	65.0	58.9	51.4	46.4	(29)
(30)		2%	MCDWB	52.5	55.0	61.5	70.0	75.0	81.8	83.0	82.1	73.7	66.1	61.3	54.0	(30)
(31)		5%	WB	42.6	43.6	49.0	54.1	60.5	66.7	68.1	68.2	63.0	57.3	49.9	44.2	(31)
(32)		5%	MCDWB	49.5	52.1	58.8	66.5	72.2	79.0	80.3	79.3	70.7	65.0	59.2	49.9	(32)
(33)		10%	WB	40.5	41.5	47.2	52.2	58.6	65.0	66.4	66.5	61.2	55.6	48.0	41.9	(33)
(34)		10%	MCDWB	45.9	48.7	55.8	63.3	69.1	75.9	77.3	76.2	68.1	61.9	54.4	46.8	(34)
(35)	Mean Daily Temperature Range		MDBR	10.9	12.4	15.5	16.9	17.0	16.7	16.9	15.7	14.8	14.3	11.7	10.4	(35)
(36)		5% DB	MCDBR	17.3	18.1	20.2	21.3	22.0	22.0	21.7	20.1	18.5	18.7	18.7	16.7	(36)
(37)			MCWBR	9.1	9.7	10.8	10.6	10.4	10.5	10.0	10.1	9.7	9.2	8.5	8.6	(37)
(38)			MCDBR	14.6	16.2	18.8	20.5	20.2	20.2	20.1	18.8	16.9	16.6	16.0	13.9	(38)
(39)		5% WB	MCWBR	8.3	9.5	10.5	11.0	10.3	10.1	9.9	10.0	9.5	9.1	8.3	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.307	0.351	0.398	0.415	0.423	0.424	0.415	0.380	0.347	0.311	0.277	(40)	
(41)		taud	2.350	2.276	2.234	2.213	2.237	2.265	2.265	2.308	2.381	2.450	2.466	2.398	(41)	
(42)		Ebn at Noon	257	271	274	271	270	268	266	263	262	255	244	245	(42)	
(43)		Edh at Noon	23	30	37	42	43	42	41	38	32	26	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	417	699	1082	1388	1558	1803	1763	1536	1167	750	435	345	(44)	
(45)		RadStd	48	89	149	208	157	166	158	151	118	84	57	49	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	N/A	N/A	N/A	+1.75	+0.72	N/A	N/A	+180	+86	(46)	
(47)	Regional (120 neighbors)	N/A	N/A	N/A	+1.22	+0.74	+0.59	N/A	N/A	+103	+27	(47)

Nomenclature: See separate page